AI SOLUTION ARCHITECT | CONSULTATIVE & ENTERPRISE THINKING

Breakout: the airline that could not get its crews back

Six questions on a real incident, 45 minutes, with a peer review inside the time.

Read this before class. Scenario: Delta Air Lines, July 2024. Nothing here is collected or marked.

What is the scenario?

On 19 July 2024 a cybersecurity company called CrowdStrike pushed out a faulty software update. It crashed Windows machines all over the world: airlines, banks, hospitals, broadcasters.

Every large US airline was hit by the same update, on the same day, on the same operating system. What happened next is the whole exercise.

Airline	How long it took to recover
American Airlines	Largely recovered by the evening of the same day, with minimal cancellations the next day.
United Airlines	About three days. Over 1,400 flights cancelled.
Delta Air Lines	More than five days. Over 7,000 flights cancelled.

Same cause, same day, same technology, three completely different outcomes. Whatever explains that gap is not the outage, because the outage was identical.

The facts you are working from

This is the whole pack. There is no second handout. Where a number you want is not here, write that it is not available rather than estimating one. Saying a number does not exist reads better than inventing it.

- Delta cancelled over 7,000 flights across five days, 19 to 25 July 2024.
- 1.3 to 1.4 million passengers were affected, per Delta's own SEC filing of 9 August 2024.
- CEO Ed Bastian put the cost at roughly $550 million: about $380m in lost revenue and $170m in expenses, partly offset by $50m in fuel savings.
- Delta's crew tracking program, the system that knows where every pilot and flight attendant currently is, was overwhelmed by the volume of changes. It came back online around 11am on 19 July and was still working through its backlog days later.
- Without reliable crew locations, Delta could not staff flights. Crew kept reaching legal flight-time limits before a full crew could be assembled, so each cancellation created more cancellations.
- Delta stated that about 60% of its mission-critical systems run on Windows, requiring a manual reset of roughly 40,000 servers.
- The US Department of Transportation's Office of Aviation Consumer Protection opened an investigation on 23 July. It was not about the outage. It was about passengers being offered future-flight credits rather than being clearly told of their right to a cash refund.

The request that has landed on your desk

"The board wants an AI system that predicts and prevents the next outage."

That sentence is your starting point, not your instruction. The first of the six questions below is whether that is actually the problem.

What you are being asked to produce (in one sentence, more below)

Six written answers, one per question. Each one between one and four sentences. Shorter is almost always stronger here. This is not a deck and not a diagram.

How the 45 minutes split: 5 minutes reading, 25 minutes writing, 10 minutes peer review, 5 minutes fixing the one thing your reviewer flagged hardest.

1. The real problem

What went wrong at Delta that did not go wrong at American?

Answer all of these. One to four sentences in total is enough.

- Say it in business words. No technology word anywhere in the sentence.
- Include one number from the pack that shows the size of it.
- Name the role inside Delta who felt it most directly. A role, not a department.

2. Who can stop you

Who can stop you, and what exactly would they stop?

Answer all of these. One to four sentences in total is enough.

- There are at least two real constraints sitting in the pack. Find both.
- One is an external body. Name it and say precisely what it governs.
- One is a legal limit. Name it and say precisely what it blocks.
- For each one, say whether you can design around it or must design within it.

3. The call

Why this and not the other thing?

Answer all of these. One to four sentences in total is enough.

- Describe two genuinely different responses to the board's request. Both must be things a reasonable person could have chosen.
- Say which one you take.

- Name the fact from the pack that decided it. Not a preference, a fact.
- Say what would have to be different for the other option to win instead.

4. The price

What did your choice cost, and who is going to argue with you about it?

Answer all of these. One to four sentences in total is enough.

- Name what you gave up, specifically. 'Money and time' is not specific.
- Name the role who loses something because of your choice.
- Say what they would say in the room when you present this.

5. The promise

What number proves it, and what happens the day you miss it?

Answer all of these. One to four sentences in total is enough.

- State a threshold with an actual digit in it.
- Say where that number is measured. Which system or report produces it.
- Say what happens when it is missed.
- Name who acts when it is missed.

6. The second home

Where else does this go?

Answer all of these. One to four sentences in total is enough.

- Give the pattern a proper name. Something another architect could look up.
- Name one other place inside the airline where the same shape appears.
- State one condition that would have to be true there, and currently is not.

Open questions, and there is no right answer to any of them

This section is not reviewed and not compared. Write what you actually think. We will take two or three of these in the room afterwards, from whoever wants to speak.

If you had been in the room when the board asked for the AI system, what is the first sentence you would say?

Which is harder where you work: finding the real problem, or being allowed to say it out loud once you have found it?

Delta's crew tracking system was not a famous piece of technology. Name one system in your own estate that nobody talks about and everything depends on.

Would your organisation rather fail less often, or recover faster? Now answer again for what it actually rewards, which may not be the same thing.

What a strong answer looks like

These are not the only right answers and your pod may land somewhere else and be perfectly defensible. They show the shape and the level of detail.
Note - You have to give your own versions. These are for reference.

Question	A strong answer
1. The real problem	"Delta could not recover. Its crew tracking system could not absorb the volume of changes, so the airline did not know where its crews were, and five days passed against American's one. The crew scheduler felt it first, because they could not assemble a legally rested crew for flights that had aircraft and passengers waiting."
2. Who can stop you	"The US Department of Transportation's Office of Aviation Consumer Protection governs how refund rights are disclosed, so any recovery plan that routes stranded passengers to future-flight credits is exposed. Separately, legal flight-time limits cap how fast any crew can be reformed no matter how good the software is. The first I can design around. The second I must design within."
3. The call	"Option A: predictive monitoring to catch a bad update before it lands. Option B: make crew tracking survive a mass re-plan, so crew state can be rebuilt in hours rather than days. I take B, because the outage was identical at American and the entire five-day gap sat in recovery. Prediction would not have moved a single number in this pack. If the pack had shown Delta's recovery was comparable and only the outage ran longer, A wins."
4. The price	"It costs the board the announcement they asked for, and it buys a programme of unglamorous work on a system nobody outside operations has heard of. The board member who wanted the AI headline argues first. The owner of the programme I am proposing to pause argues second, and more effectively."
5. The promise	"Crew state is fully reconstructed within four hours of a mass disruption, measured by the crew tracking system's own backlog queue returning to normal depth. If a scheduled recovery exercise misses four hours, the Chief Operating Officer holds the next phase of the programme until the gap is closed."

6. The second home	"Bounded recovery time as a design constraint, sometimes written as a recovery-time objective. The same shape appears in baggage reconciliation after a mass reroute. To transfer it there, baggage would need what crew needs: one authoritative state store that can be rebuilt from an event log rather than reassembled from downstream systems. Today it does not have one."

What to avoid

Trap	Why it costs you
Answering question 1 with the outage	"A CrowdStrike update crashed the systems" describes the event, not the problem. American had the same event and recovered in a day. If your answer is also true of American, it is not Delta's problem.
Naming a body by its function	"The regulator" and "compliance" are functions. Name the body on the door, and say what it actually governs. Here, note that the DOT investigation was about refund disclosure, not about the outage.
A second option nobody would pick	If the rejected option is described so that no reasonable person would choose it, it was never an option. It was a straw man wearing one.
A choice that cost nothing	If every clause in your answer to question 4 names a gain, you have described a wish rather than a decision.
A threshold with no digit	"Significantly faster" and "as quickly as possible" are not thresholds. The number can be borrowed from the pack or chosen by you. It just has to exist.
A promise nobody owns	If nothing happens when the number is missed, and nobody is named, it is an assertion rather than a promise.
"Reusable", "scalable", "extensible"	Adjectives, not patterns. A pattern has a name another architect could look up. A precondition that excludes nothing is decoration.
Starting with what you would build	Everyone in the room already has an opinion about what Delta should build. Park it for five minutes and answer question 1 honestly, or every answer after it inherits the guess.

What your two-minute report should sound like

This is what you handover in the main room (The 6 discussed above, in a sentence format)

Three sentences, in this order, from whoever your pod nominated. No preamble and no apology for what you did not finish. This is the Debrief, the one which would be discussed pod wise.

e.g. *"The real problem was that Delta could not recover, not that it went down: five days against American's one, and the crew scheduler could not build a legal crew. So we chose to harden crew-state recovery rather than build outage prediction, because the outage was identical at American. It costs the board the AI announcement they asked for, and the programme owner we are pausing will say so."*

That is the whole report. If you can say it in under two minutes without reading, your six answers are in good shape.

Peer review, inside the 45 minutes (Optional - If you have time left)

Pod 1 with Pod 2, Pod 3 with Pod 4, Pod 5 with Pod 6. Ten minutes. Read theirs against these five checks, be specific, and be quick.

Check	Flag it if
Technology in answer 1	Any technology word appears in their problem statement. This is the most common fault.
A named body	They wrote a function rather than the name of a body, or they missed one of the two constraints.
Two real options	The rejected option is a straw man, or there is only one option described twice.
A real price	The cost is generic, or no named role loses anything.
A digit and an owner	The promise has no number, no source, no consequence, or nobody who acts.

The one thing my reviewer flagged hardest, and what I changed because of it

Your pod

Pod number	Who is presenting	Who reviewed my work	Timekeeper

House rules for the 45 minutes

- Read this document before class. The 5 minutes of reading time inside the session is a reminder, not a first read.
- Nothing here is collected or marked. It is practice, and the practice is the point.
- Your pod reasons together. Each of you writes your own answers. Identical documents help nobody.
- Peer review happens inside the 45 minutes, not after.
- Decide your presenter and timekeeper before the clock starts. Running out of time while deciding is the most common way pods overrun.
- Where a number is not in the pack, write that it is not available. Do not estimate one.